

Machine-Learning Interatomic Potentials for Probing Adsorbate

Diffusion, Desorption and Sticking Kinetics

Saurabh Sivakumar, Trevor Price, Sudheesh Kumar Ethirajan & Ambarish Kulkarni

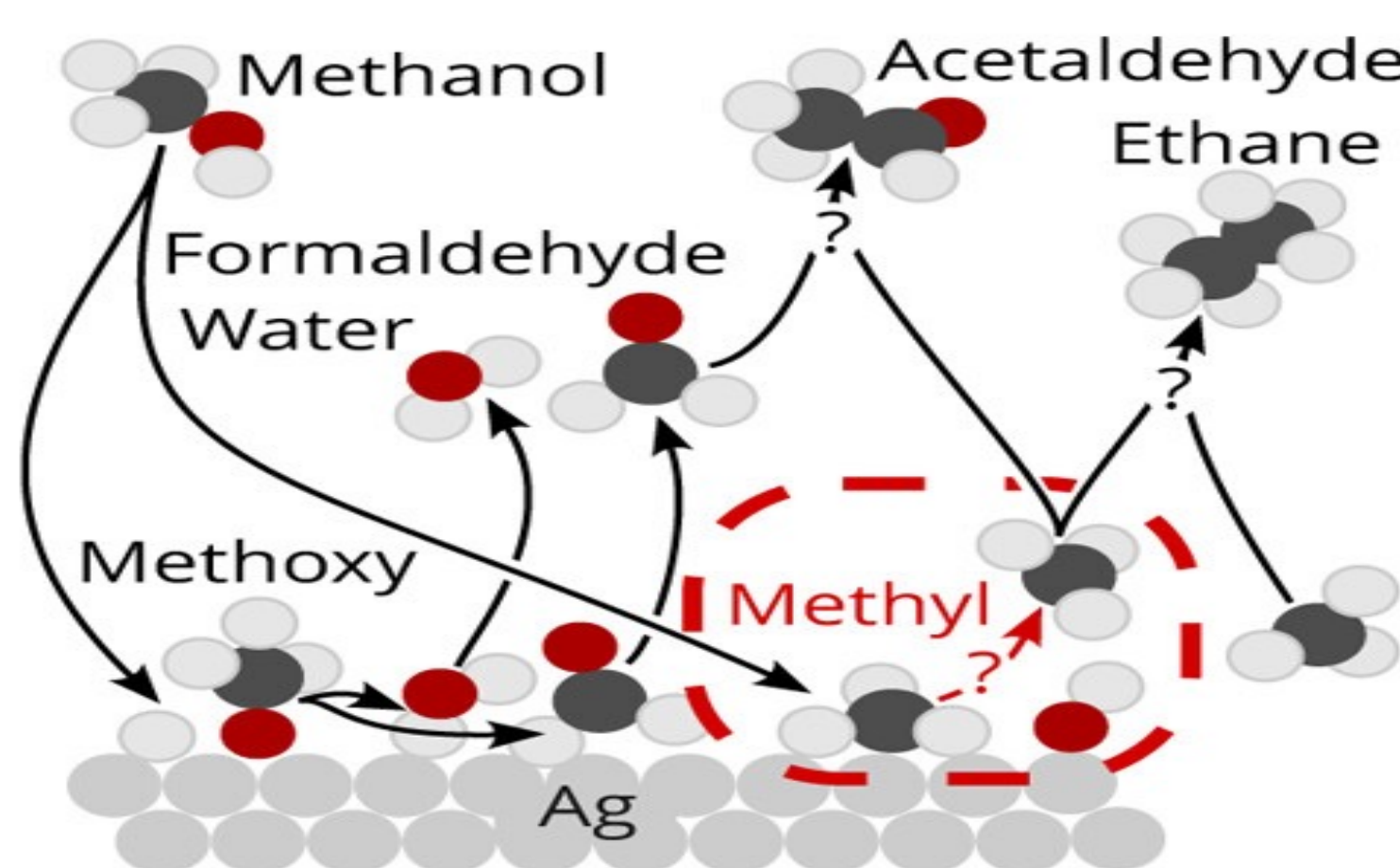
Department of Chemical Engineering, University of California, Davis



Motivation & System

Problem - Coupled gas phase and surface mechanisms

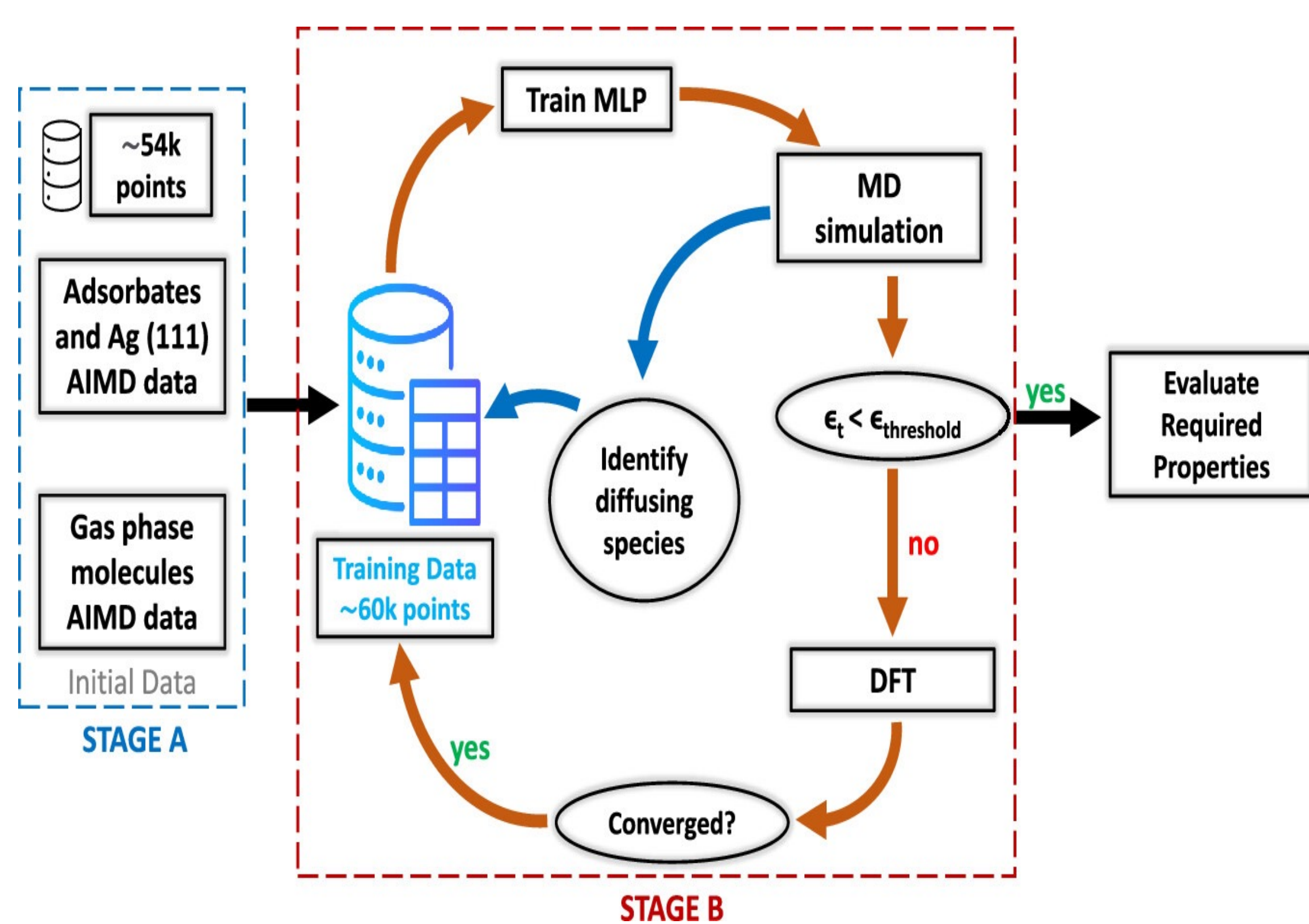
- Reactions like OCM and Partial Oxidation of Methanol
- Several elementary steps involving gas and surface intermediates
- Reaction networks are usually extremely complex



Partial Oxidation of Methanol over silver:

- Previously well studied mechanism in 2016 – 53 reactions
- Experimental Collaborators identified 3 relevant intermediates acetaldehyde (CH_3CHO), dimethyl ether (CH_3OCH_3) and methyl formate (HCOOCH_3) in 2021

Methods

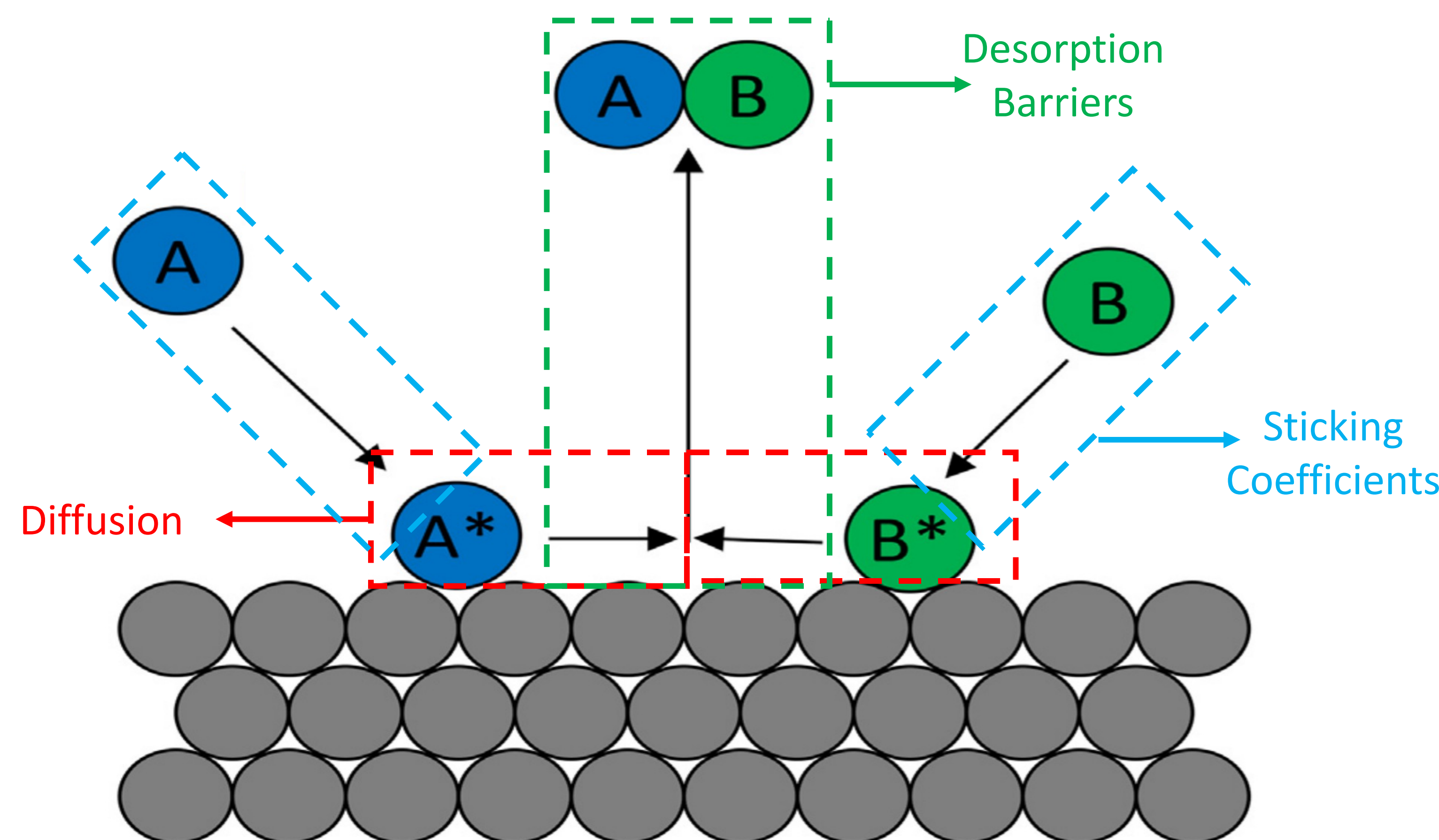


Active learning used to train a DeepPot-SE MLIP on over 670 gas phase intermediates and adsorbates on an Ag (111) surface

MLIP Architecture	MAE Energy (eV)	MAE Forces (eV/Å)	2 ns MD Simulation Time (h)
DeepPot-SE	0.12	0.04	~2.2
Equiformer-v2	0.05	0.03	N/A
MACE - Small	0.11	0.05	~15.3
MACE - Medium	0.08	0.04	~19.6
MACE - Large	0.07	0.04	~23.5

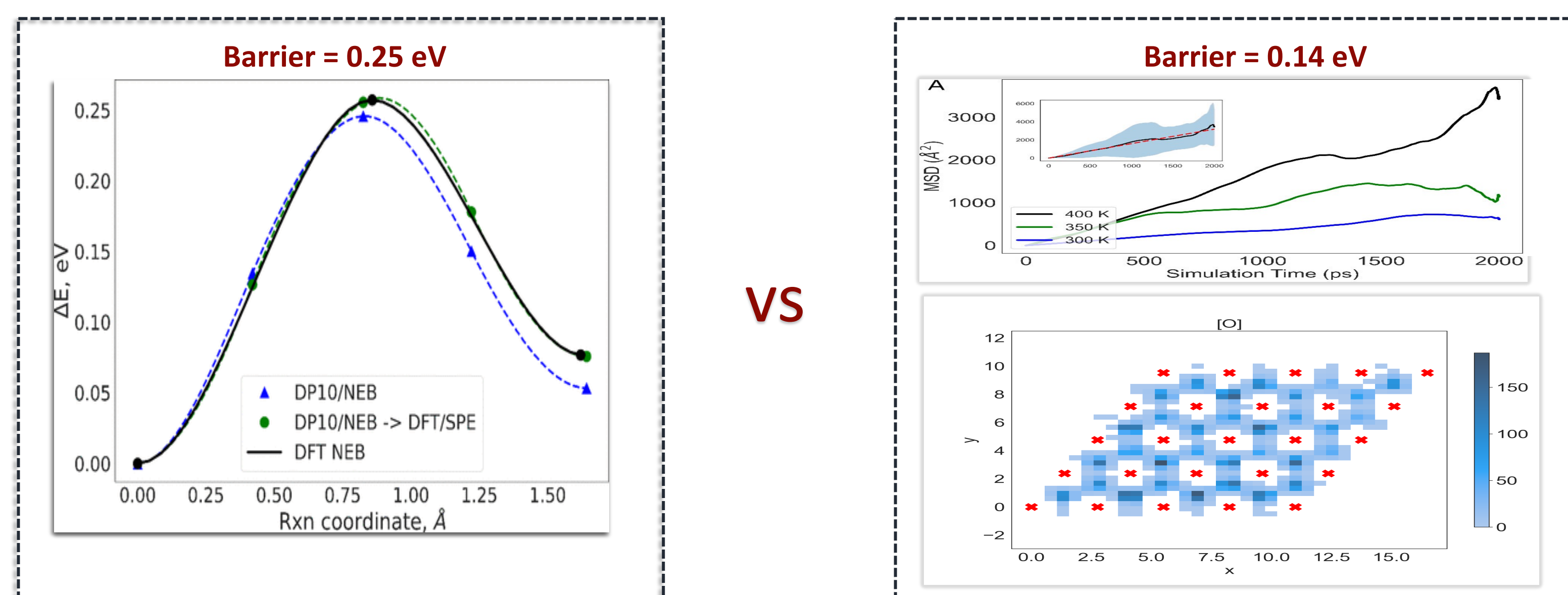
* MD (NVT) simulation time measured for a 68 atom system on a single H100 GPU with LAMMPS compiled with Kokkos

Reaction sub-steps



Subprocesses have significant impact on reaction mechanisms and rates. Mechanistic insight necessary for tuning catalysts

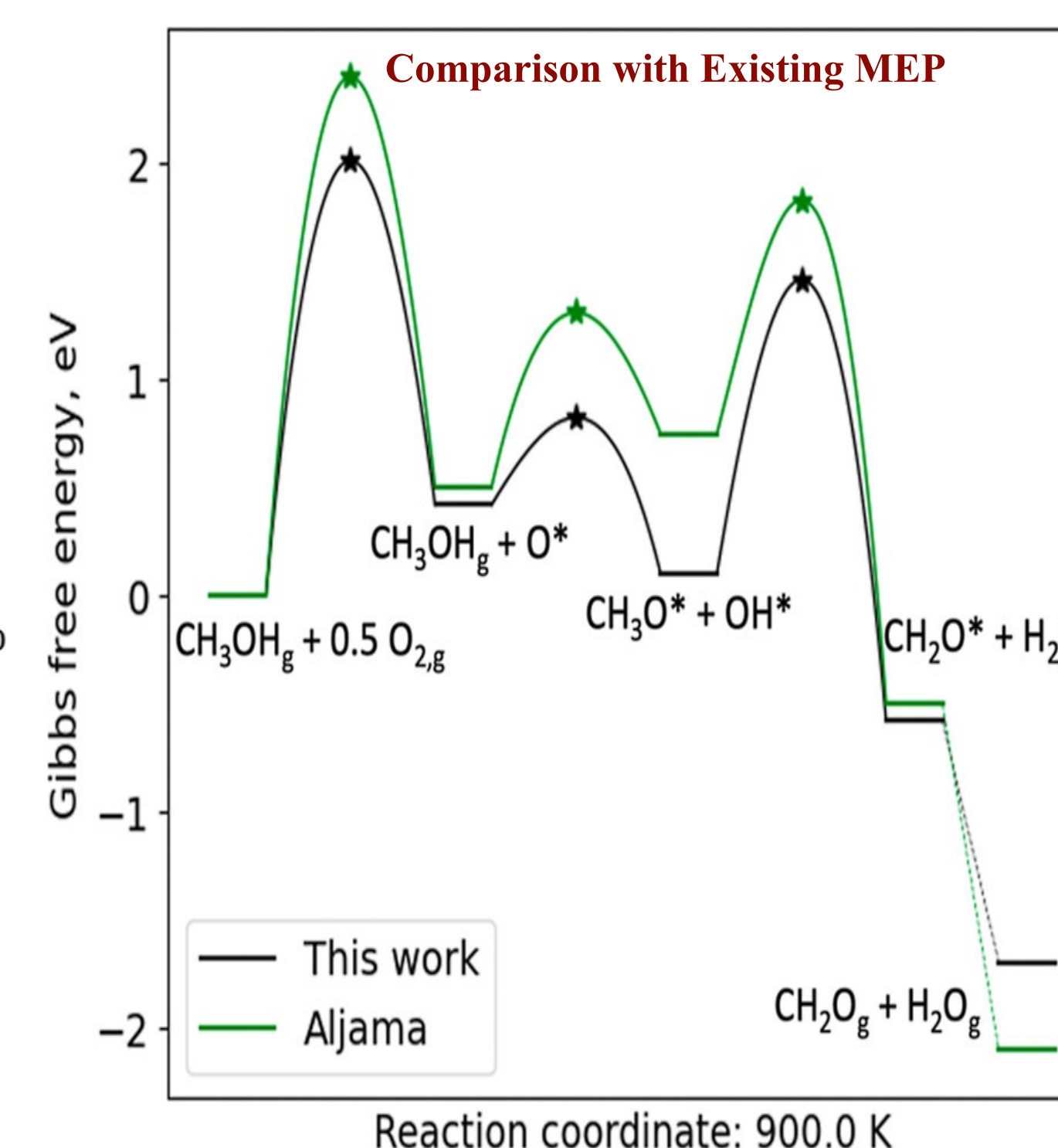
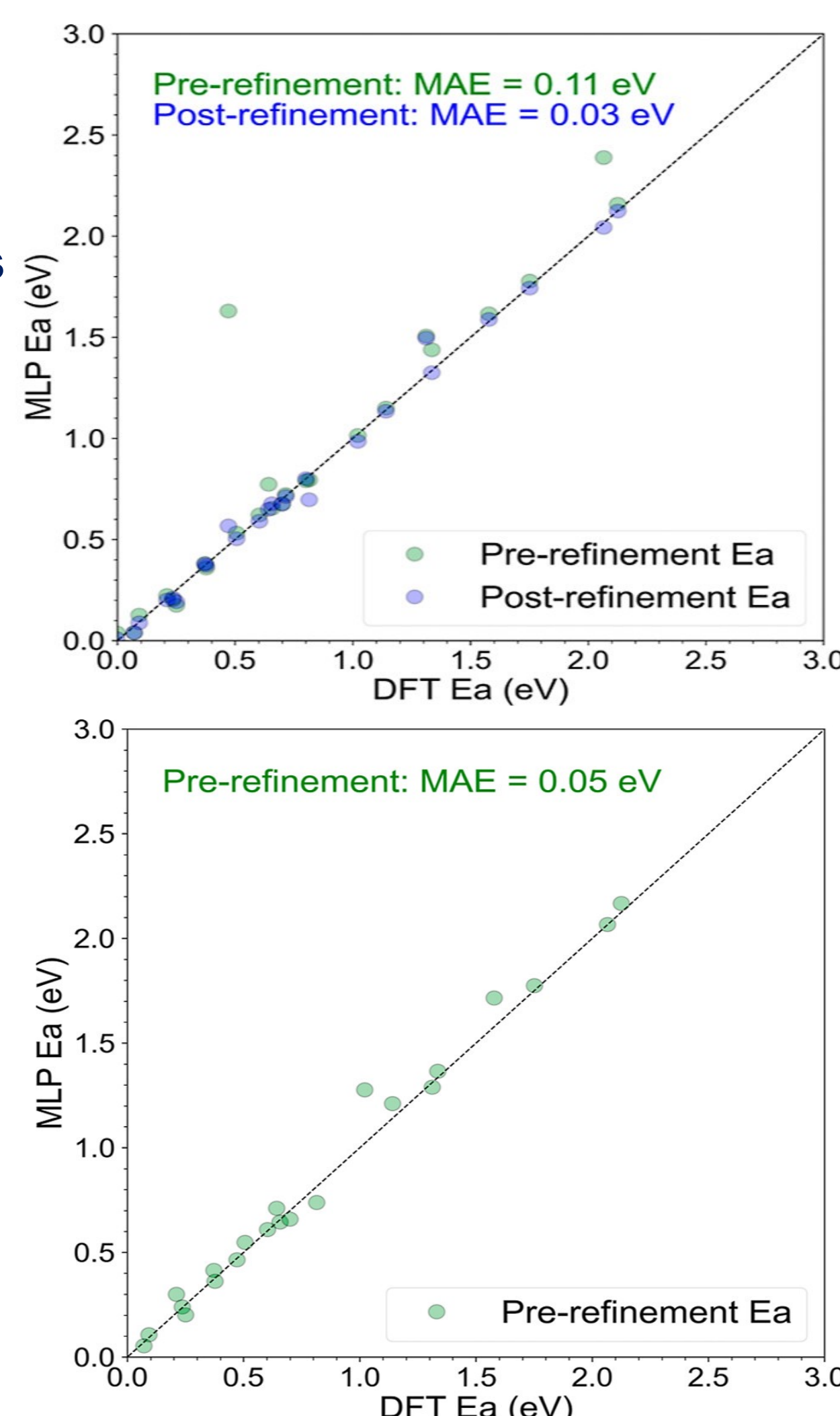
Diffusion & Reactive Events



Diffusion barriers and diffusivity computed from MD simulations contrasted with a harmonic NEB calculations for Oxygen diffusion

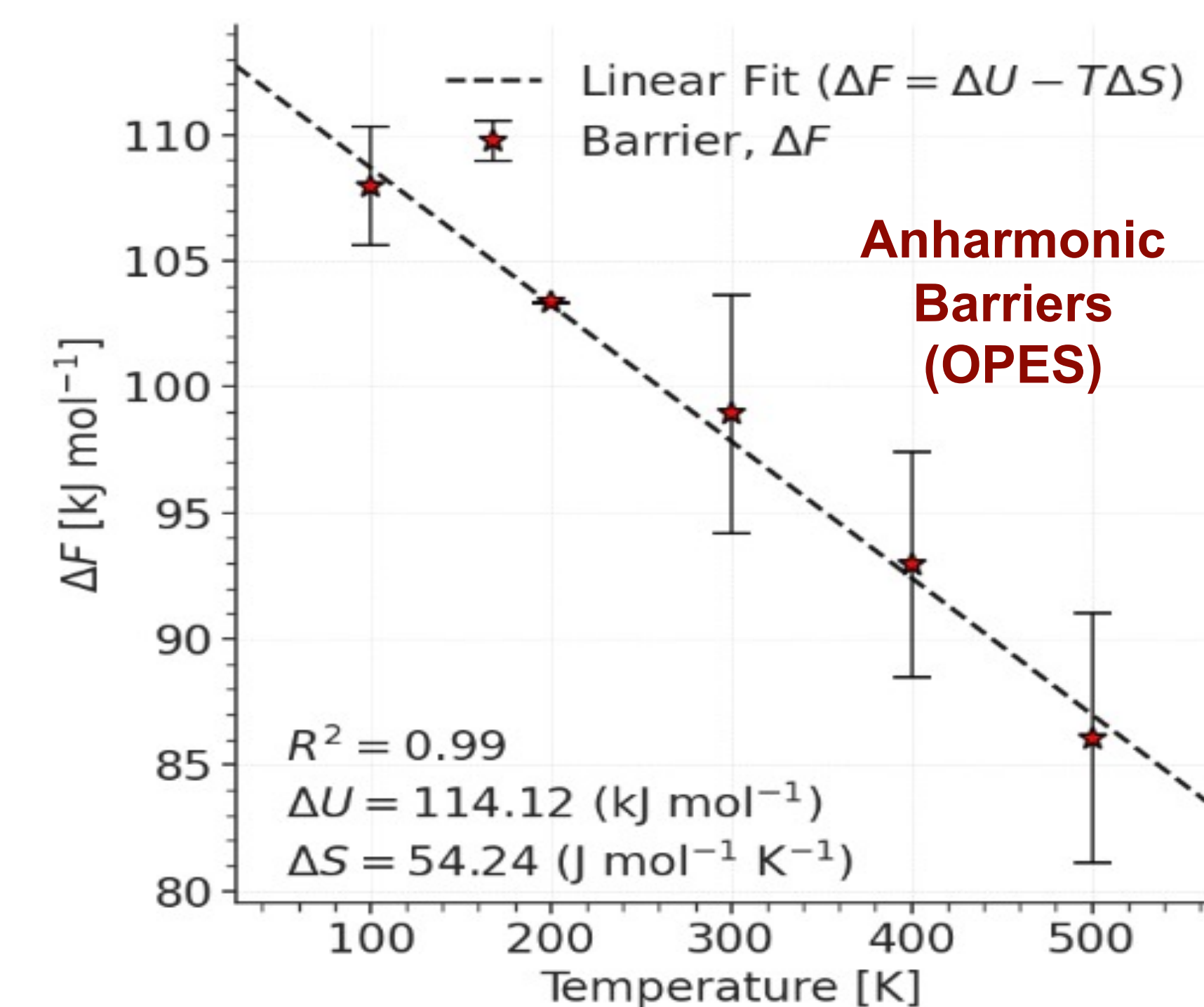
DeepPot-SE + Reaction class – Identified 53/53 TS (7x speed over DFT)

Pre-Trained Equiformer v2 – Identified 47/53 TS (20x speed over DFT)



MLIPs integrated with Transition state workflows to enable speedup and mechanism discovery

Desorption & Beyond



- Desorption kinetics at higher temperatures is characterized by Large-Amplitude Motions which the harmonic approximation fails to capture sufficiently
- Desorption barriers obtained for Methyl through enhanced sampling to compare and validate our MKM model with experimental measurements for methyl (a radical that can be detected and measured)
- Computation of sticking coefficients through MD
- MKM flux diagrams + sensitivity analysis reveal O2 dissociation as the rate-limiting step

Challenges & Future Work

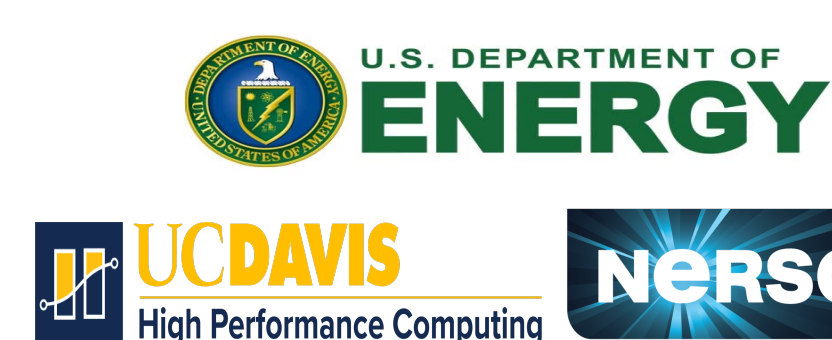
Challenges

- Surface and system complexity - Coverage dependence, Dispersion effects, surface reconstruction
- Catalyst Support and its effect on reaction
- Formation of oxide layer and/or coking
- Facet-specific behavior (e.g., 211)

Future Work

- Extending to other facets that are likely more active
- Surface coverage dependent properties

Acknowledgements



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References

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Contact

Email: sausiva@ucdavis.edu

